Figure 9.3 Use case diagram for RES4 asset token Dapp

9.2.2 Contract diagram

The contract diagram expands on the guideline provided by the use case diagram and adds more elements of design: data structures, modifiers, events, and functions (function headers). The contract diagram has only three elements: data, events, and functions. The access rules are specified inside the functions with `require` statements (`require (condition);`), not at the header of the functions. The RES4 follows the ERC721 standard that defines a set of function headers.

Figure 9.4 shows the contract diagram for RES4. Besides the data and the events, the RES4 functions in the contract diagram follow the use case diagram. These functions are `addAsset()`, `build()`, `approve()`, and `transfer()`. The functions `appreciate()` and `depreciate()` are for supporting the operations of the assessor role. These operations allow town officials to increase or decrease the current value of an asset. The events specified in the contract diagram are required by the ERC721 standard.